

David Brandão

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EDUCATION

Instituto Superior de Engenharia do Porto (Current average: 15/20)
Bachelor's in Informatics Engineering

Sep 2024 – Jun 2027
Porto, Portugal

PROFESSIONAL EXPERIENCE

ROQ Group

Full-Stack Developer Intern

Oct 2024 – Feb 2025
V.N. Famalicão

- Developed and extended internal business applications using .NET, implementing backend business logic and frontend interfaces based on engineering requirements.
- Built management features integrating existing business data, including user/article workflows, SQL queries and supporting database structures.

CERTIFICATIONS & EXAMS

Cisco Certified Network Associate (CCNA)

Aug 2026

AWS Solutions Architect - Associate (SAA-C03): In Progress

Scheduled Oct 2026

PROJECTS

Risk Assistant: Threat Analysis Platform

AWS, Terraform, Python, Amazon Bedrock

- Designed the AWS backend for a 48-hour hackathon team project: a serverless, event-driven threat analysis platform for URLs, messages and images using API Gateway, Cognito, Lambda, SQS, DynamoDB, S3 and Amazon Bedrock.
- Built asynchronous SQS-based processing with DLQ, content caching and AI fallback logic; authored a prioritized architecture review (fail-safe defaults, job recoverability, IAM hardening) documenting production gaps and trade-offs.

Serverless Runtime Performance Benchmark: Rust vs Java 21

AWS Lambda, Rust, Java 21, Terraform, CloudWatch, SnapStart

- Designed an AWS Lambda benchmark across 2,000+ invocations per configuration, with automated invocation and log-analysis tooling comparing Rust, Java 21 baseline and Java 21 + SnapStart under functionally equivalent image-processing workloads.
- Measured ~10× lower cold-start latency, ~78% lower memory usage and ~81% lower billed duration with Rust; found SnapStart underperformed baseline Java in this workload, contrary to initial expectations.

Automated Multi-Site Network Infrastructure

Cisco IOS, Ansible, OSPF, IPsec

- Designed a three-site network using OSPF over GRE/IPsec VPNs, VLANs, inter-VLAN routing with HSRP, NAT, ACLs and DHCP; deployed and validated the topology entirely on physical Cisco hardware.
- Built Ansible tooling for automated config backups and operational checks, with documentation that distinguishes configured intent from verified live state.

SKILLS

Cloud & AWS: EC2, S3, VPC, IAM, Lambda, API Gateway, SQS/SNS, RDS, DynamoDB, Cognito, CloudWatch, Bedrock

Networking: TCP/IP, IPv4/IPv6, VLANs, STP, EtherChannel, OSPF, NAT, ACLs, DHCP, WLAN

Infrastructure & Automation: Terraform, Ansible, Docker, Linux, Bash, Git

Programming: Python, Java, C, C#, SQL

LANGUAGES

Portuguese — Native

English — C1 Advanced (CEFR) | EF SET Certificate